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Questions

Q1.

Ammonia is manufactured in the Haber Process.

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

In this process, pressures between 100 atm and 300 atm and temperatures between 675 K and 725 K are usually used.

(a) (i) Calculate $\Delta S^{\ominus}_{system}$ for this reaction at 298 K.
Include a sign and units with your answer.

[Standard molar entropy, $S^{\ominus}$, for 1 mol of each gas, in J K^{-1} mol^{-1}: $N_2 = 191.6$ $H_2 = 130.6$ $NH_3 = 192.3$]

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(ii) Explain, using ideas about disorder, whether the sign of your answer to (a)(i) is as expected.

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(iii) At 700 K, the enthalpy change for this reaction is $\Delta H = -110.2$ kJ mol^{-1}.
Calculate the entropy change of the surroundings, $\Delta S_{surroundings}$, at 700 K. Include a sign and units in your answer.

(2)

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(iv) At 700 K, the total entropy change, $\Delta S_{total} = -78.7\ J\ K^{-1}\ mol^{-1}$.
Calculate ΔS_{system} for this reaction at 700 K. Include a sign and units in your answer.

(1)

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*(v) Explain **one** advantage and **one** disadvantage of using a temperature higher than 700 K in the Haber process.
Consider the effect of an increase in temperature on the rate of reaction, on the values of $\Delta S_{surroundings}$ and ΔS_{total} and on the equilibrium constant.

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(b) (i) Write the expression for the equilibrium constant, K_p , for the reaction in the Haber Process.

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

(1)

(ii) A mixture of nitrogen, hydrogen and ammonia is at equilibrium at 255 atm and 700 K. The partial pressure of nitrogen is 25 atm and that of hydrogen is 150 atm.
Calculate the equilibrium constant, K_p , for the formation of ammonia, at 700 K. Include units in your answer.

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(iii) The pressure of the system is doubled at constant temperature.
Explain the effect of this on the yield of ammonia by considering the change in the K_p expression and hence the position of equilibrium.

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(Total for question = 20 marks)

Q2.

Equal amounts of W and X are mixed and allowed to reach equilibrium.

$$W + X \rightleftharpoons Y + Z$$

The value of the equilibrium constant is $K_c = 4.85$. At equilibrium, the mixture will contain

- **A** almost all Y and Z
- **B** almost all W and X
- **C** W, X, Y and Z but there is less Y and Z than W and X
- **D** W, X, Y and Z but there is more Y and Z than W and X

(Total for question = 1 mark)

Q3.

A student carried out an experiment to determine the equilibrium constant for the reaction to form ethyl ethanoate from ethanol and ethanoic acid.

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$$CH_3COOH + CH_3CH_2OH \rightleftharpoons CH_3COOCH_2CH_3 + H_2O$$

Different amounts of each substance were added to conical flasks, each containing 2.0 cm^3 of 1.0 mol dm^{-3} hydrochloric acid.

Conical flask	Volume added / cm^3				
	HCl (aq)	H_2O (l)	CH_3COOH (l)	$CH_3COOCH_2CH_3$ (l)	CH_3CH_2OH (l)
1	2.0	1.0	0	2.0	0
2	2.0	0	5.0	0	5.0
3	2.0	0	4.0	0	3.0
4	2.0	0	0	3.0	0

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The flasks were then stoppered and left for a week to reach equilibrium.

Each mixture was then titrated with 1.0 mol dm^{-3} sodium hydroxide.

(a) In which flask(s) was the equilibrium approached from the right-hand side of the equation?

(1)

- ☒ **A** flask 1 only
- ☒ **B** flask 4 only
- ☒ **C** flasks 1 and 4 only
- ☒ **D** flasks 2 and 3 only

(b) Which of these statements explains why it is possible to titrate the reaction mixture directly to find the equilibrium concentrations?

(1)

- ☒ **A** the equilibrium reaction is slow
- ☒ **B** the sodium hydroxide quickly hydrolyses the ester
- ☒ **C** all the reactant concentrations remain constant during the titration
- ☒ **D** a buffer solution forms in the reaction

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(Total for question = 2 marks)

Q4.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The equation for the equilibrium decomposition of hydrogen iodide into hydrogen and iodine can be written in two ways.

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$2HI(g) \rightleftharpoons H_2(g) + I_2(g)$ Equilibrium constant = K_c

or

$HI(g) \rightleftharpoons \frac{1}{2}H_2(g) + \frac{1}{2}I_2(g)$ Equilibrium constant = K_c'

What is the expression linking the two equilibrium constants?

- ☐ **A** $K_c = (K_c')^2$
- ☐ **B** $K_c = K_c'$
- ☐ **C** $K_c = 2(K_c')$
- ☐ **D** $K_c = \sqrt{K_c'}$

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(Total for question = 1 mark)

Q5.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A homogeneous equilibrium is shown.

$$W + X \rightleftharpoons 2Y + Z$$

What is the K_c expression for this equilibrium?

- ☐ **A** $K_c = \frac{2[Y][Z]}{[W][X]}$
- ☐ **B** $K_c = \frac{[Y]^2[Z]}{[W][X]}$
- ☐ **C** $K_c = \frac{[W][X]}{2[Y][Z]}$
- ☐ **D** $K_c = \frac{[W][X]}{[Y]^2[Z]}$

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(Total for question = 1 mark)

Q6.

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This question is about halides.

(a) Silicon tetrachloride, $SiCl_4$, is used in the manufacture of optical fibres. It can be made by the reaction of silicon carbide, SiC, with hydrogen chloride using a catalyst of nickel(II) chloride.

$$SiC(s) + 4HCl(g) \rightarrow SiCl_4(l) + CH_4(g) \qquad \Delta H = -631.3\,kJ\,mol^{-1}$$

The standard molar entropies $S^{\ominus}$, of the substances are shown in the table.

Substance	SiC(s)	HCl(g)	$SiCl_4(l)$	$CH_4(g)$
$S^{\ominus}/JK^{-1}\,mol^{-1}$	+16.5	+186.8	+239.7	+186.2

(i) Calculate the total entropy change, ΔS_{total}, for this reaction, at 298 K, using the information given. Include a sign and units in your answer which should be given to an appropriate number of significant figures.

(5)

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(ii) In industry, the reaction is carried out at 700 °C.
By considering entropy and other relevant factors, justify the use of this temperature.

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(iii) Use your answer from (a)(i) to calculate the equilibrium constant for the reaction at 298 K.

(2)

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(b) The following data can be used in a Born-Haber cycle for calcium iodide, CaI_2.

Letter	Energy change	Value / kJ mol^{-1}
A	Enthalpy change of atomisation of calcium	+178.2
B	First ionisation energy of calcium	+590
C	Second ionisation energy of calcium	+1145
D	Enthalpy change of atomisation of iodine ($\frac{1}{2}I_2(s) \rightarrow I(g)$)	+106.8
E	Electron affinity of iodine	−295.4
F	Lattice energy of calcium iodide	−2074

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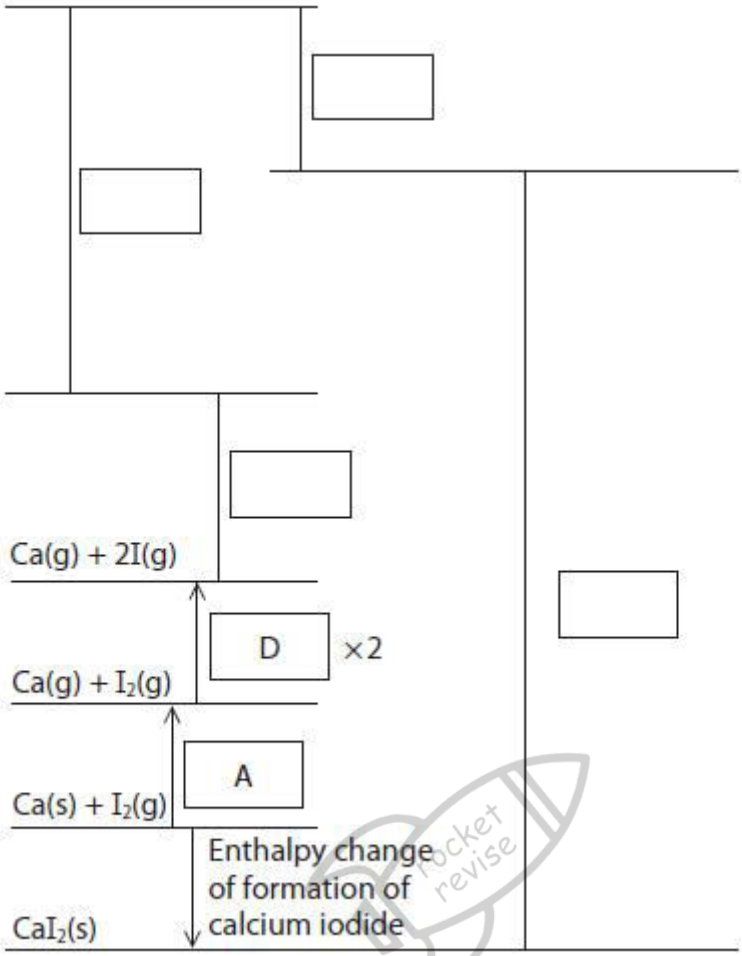
(i) Complete the Born-Haber cycle by adding letters in the boxes for the energy changes, relevant species on the blank lines and arrowheads to show the direction of each energy change.

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(ii) Use the data to calculate the value for the enthalpy change of formation of calcium iodide.

(2)

(iii) The value for the lattice energy of calcium iodide determined experimentally by using the Born-Haber cycle differs significantly from the theoretical calculated value.

Explain why the Born-Haber and the theoretical values for the lattice energies are similar for calcium fluoride but significantly different for calcium iodide.

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(c) The electron affinity of iodine is –295.4 kJ mol^{-1}.

Explain how the electron affinity of chlorine differs from that of iodine.

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(Total for question = 22 marks)

Q7.

The reversible reaction between hydrogen chloride and oxygen produces water **vapour** and chlorine.

$$4HCl(g) + O_2(g) \rightleftharpoons 2H_2O(g) + 2Cl_2(g) \qquad \Delta H = -114 \text{ kJ mol}^{-1}$$

(a) Explain what effect, if any, each of the following changes has on the yield of chlorine at equilibrium **and** on the equilibrium constant, K_p .

(i) An increase in the total pressure

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(ii) An increase in the temperature

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(iii) The use of a catalyst

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(b) 0.850 mol of hydrogen chloride was mixed with 0.600 mol of oxygen and allowed to reach equilibrium in a closed flask.
At equilibrium the total pressure was 1.50 atm and there was 0.250 mol of chlorine in the flask.

$$4HCl(g) + O_2(g) \rightleftharpoons 2H_2O(g) + 2Cl_2(g)$$

(i) Complete the table.

(3)

Substance	Initial amount / mol	Equilibrium amount / mol	Mole fraction at equilibrium
HCl	0.850		
O_2	0.600		
H_2O	0		
Cl_2	0	0.250	0.189
Total moles at equilibrium =			

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(ii) Write the expression for the equilibrium constant, K_p .

(1)

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(iii) Use your answers to (b)(i) and (b)(ii) to calculate the value for K_p. Give your answer to an appropriate number of significant figures, and include units.

(3)

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(iv) Use your answer to (b)(iii) to calculate a value for the total entropy change of the reaction, ΔS_{total}.

(2)

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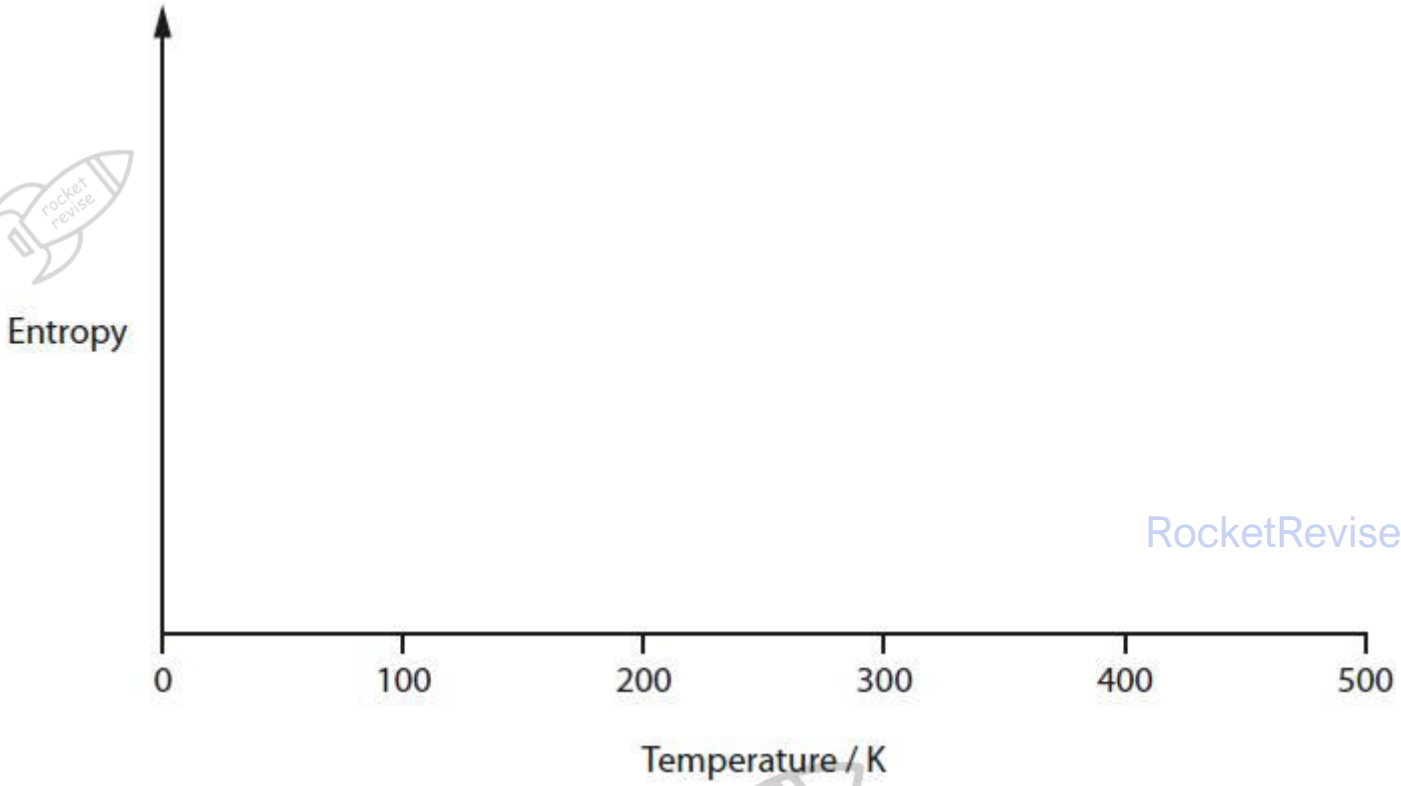
(c) Draw a sketch of entropy against temperature for water to illustrate the entropy changes as temperature increases, including when water changes state.

A scale is not required for the vertical axis

(3)

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(Total for question = 19 marks)

Q8.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

The total entropy change, ΔS_{total} , of a reaction at 298 K is −85.0 J K^{-1} mol^{-1}.

What is the value of the equilibrium constant for this reaction at 298 K?

[R = 8.31 J mol^{-1} K^{-1}]

☐ **A** 3.61×10^{-5}

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☐ **B** 9.07×10^{-1}

☐ **C** 9.66×10^{-1}

☐ **D** 2.77×10^{4}

(Total for question = 1 mark)

Q9.

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Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The water gas reaction is used in the manufacture of hydrogen.

$$C(s) + H_2O(g) \leftrightarrow CO(g) + H_2(g) \qquad \Delta H^{\ominus} = +131.2 \text{ kJ mol}^{-1}$$

(a) What is the equilibrium constant, K_c, for this reaction?

(1)

☐ **A** $K_c = [CO][H_2]$

☐ **B** $K_c = \dfrac{[CO][H_2]}{[C]}$

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☐ **C** $K_c = \dfrac{[CO][H_2]}{[H_2O]}$

☐ **D** $K_c = \dfrac{[CO][H_2]}{[H_2O][C]}$

(b) What happens to the equilibrium constants of the forward and reverse reactions when the temperature is **increased**?

(1)

	K_c of forward reaction	K_c of reverse reaction
☐ **A**	increases	increases
☐ **B**	increases	decreases
☐ **C**	decreases	increases
☐ **D**	decreases	decreases

(Total for question = 2 marks)

Q10.

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Ethanol may be manufactured by the hydration of ethene.

$$C_2H_4(g) + H_2O(g) \rightleftharpoons C_2H_5OH(g) \qquad \Delta H^{\ominus} = -45.3 \text{ kJ mol}^{-1}$$

(a) In a laboratory investigation of this reaction, 1.00 mol of ethene was mixed with 1.00 mol of steam at 150 °C. At equilibrium, when the total pressure of the system was 50.0 atm, 0.450 mol of ethanol had formed.

(i) Give the expression for the equilibrium constant, K_p, for the reaction.

(1)

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(ii) Calculate the equilibrium constant, K_p, for the hydration of ethene at 150 °C.
Include units with your answer.

(5)

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(b) The manufacture of ethanol is carried out at 230 °C and 70 atm; the overall conversion into ethanol is 95 %.

Comment on these conditions in relation to their effect on the equilibrium and the overall yield of ethanol.

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(Total for question = 9 marks)

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Q11.

For each question, select one answer from A to D and put a cross in the box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The reaction shown is at equilibrium. The forward reaction is endothermic.

$$C(s) + CO_2(g) \leftrightarrow 2CO(g)$$

(a) Which will **increase** when the temperature is lowered?

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(1)

- ☐ **A** the mole fraction of carbon dioxide
- ☐ **B** the partial pressure of carbon monoxide
- ☐ **C** the rate of the backward reaction
- ☐ **D** the value of K_p for the forward reaction

(b) At 680 °C and 1 atm, 52.6 % of the molecules in the gas mixture are carbon monoxide. What is the partial pressure of carbon dioxide, in atmospheres?

(1)

- ☐ **A** 0.237
- ☐ **B** 0.263
- ☐ **C** 0.474
- ☐ **D** 0.526

(Total for question = 2 marks)

Q12.

For each question, select one answer from A to D and put a cross in the box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The equation for the precipitation of lead(II) chloride is shown.

$$Pb^{2+}(aq) + 2Cl^-(aq) \leftrightarrow PbCl_2(s)$$

What are the units of the equilibrium constant, K_c?

- ☐ **A** $dm^9\ mol^{-3}$
- ☐ **B** $dm^6\ mol^{-2}$
- ☐ **C** $mol^2\ dm^{-6}$
- ☐ **D** $mol^3\ dm^{-9}$

(Total for question = 1 mark)

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Q13.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

This question is about the equilibrium system used in the industrial production of nitric acid.

$$4NH_3(g) + 5O_2(g) \Rightarrow 4NO(g) + 6H_2O(g) \qquad \Delta H = -905 \text{ kJ mol}^{-1}$$

The process is carried out at a temperature of 800°C, a total pressure of 10 atm and in the presence of a platinum-rhodium catalyst.

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(a) Which conditions of temperature and pressure would be expected to give the greatest equilibrium yield of NO?

(1)

- ☐ **A** high temperature and high pressure
- ☐ **B** high temperature and low pressure
- ☐ **C** low temperature and high pressure
- ☐ **D** low temperature and low pressure

(b) Which of these statements **best** explains why only a small amount of energy is required to maintain this process?

(1)

- ☐ **A** the catalyst increases the equilibrium yield of NO
- ☐ **B** the catalyst decreases the activation energy
- ☐ **C** the reaction is thermodynamically feasible
- ☐ **D** the reaction is highly exothermic

(c) What are the units of the equilibrium constant, K_p, for this reaction?

(1)

- ☐ **A** atm^{-1}
- ☐ **B** atm
- ☐ **C** $dm^3\ mol^{-1}$
- ☐ **D** $mol\ dm^{-3}$

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(Total for question = 3 marks)

Use this space for any rough working. Anything you write in this space will gain no credit.

Q14.

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This question is about the homogeneous system

$$H_2(g) + I_2(g) \leftrightarrow 2HI(g)$$

(a) Write the expression for the equilibrium constant, K_c , of this system.

(1)

(b) Chemist **A** carried out an experiment to determine K_c.

0.00500 mol of H_2 and 0.0100 mol of I_2 were heated in a sealed container of volume 1.00 dm^3.
At equilibrium, the mixture was found to contain 0.00968 mol of HI.
Calculate K_c , using these data and your expression from (a).
Include units, if any.
Give your answer to an appropriate number of significant figures.

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(4)

(c) Chemist **B** determined the value of K_c at five different temperatures.

Temperature, T/K	$1/T$ (K^{-1})	K_c	ln K_c
523		122	4.80
577	0.00173	99.2	
620	0.00161	86.5	4.46
669	0.00149	75.6	4.33
721	0.00139	66.8	4.20

(i) Complete the table of data.

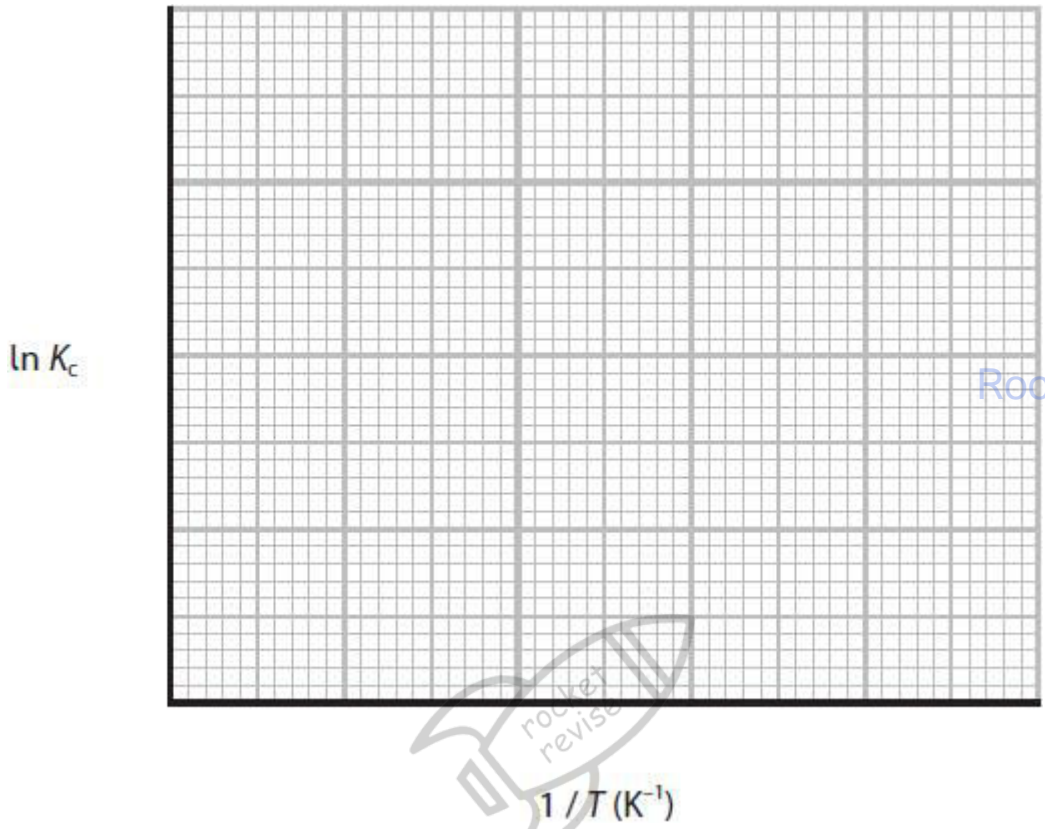
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(2)

(ii) Plot a graph of ln K_c against 1 / T.
Include a line of best fit.

(3)



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(iii) Determine the gradient of your graph.
You **must** show your working on the graph.

(1)

(iv) The gradient of the graph from (c)(iii) is related to the enthalpy change of the reaction, ΔH.

$$\text{gradient} = -\frac{\Delta H}{R}$$

Calculate ΔH, in J mol^{-1}, using your value for the gradient.

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(2)

(v) The free energy change for the reaction, ΔG, is related to entropy and can be calculated from K_c .

$$\Delta G = -RT \ln K_c$$

Calculate ΔG at **620 K**, using this equation and data from the table.
Include units.

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(2)

(vi) The equation relating ΔG, ΔH and ΔS_{system} is shown.

$$\Delta G = \Delta H - T\Delta S_{system}$$

Calculate ΔS_{system} at 620 K, using your answers to (c)(iv) and (c)(v).

(2)

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(d) Explain, in terms of the total entropy change, ΔS_{total} , why this reaction is feasible at any temperature. Assume that both ΔH and ΔS_{system} are independent of temperature.

(3)

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(Total for question = 20 marks)

Q15.

Answer the question with a cross in the box you think is correct ☒. If you change your mind

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about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

What are the units of K_p for the equilibrium shown?

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

- ☐ **A** atm^{-1}
- ☐ **B** atm
- ☐ **C** atm^{-2}
- ☐ **D** atm^2

(Total for question = 1 mark)

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Q16.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The reaction shown occurs at 360°C and 1 atm.

$$CO + H_2O \rightleftharpoons CO_2 + H_2 \qquad \Delta H = -41.2\,kJ\,mol^{-1}$$

What is the type of equilibrium and how is K_c affected by an **increase** in temperature?

	Type of equilibrium	Effect of increasing temperature on K_c
☐ **A**	heterogeneous	decreases
☐ **B**	homogeneous	decreases
☐ **C**	heterogeneous	increases
☐ **D**	homogeneous	increases

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(Total for question = 1 mark)

Q17.

This question is about the reaction between nitrogen monoxide and oxygen.

$$NO(g) + \tfrac{1}{2}O_2(g) \rightleftharpoons NO_2(g)$$

(a) The results of a series of kinetics experiments are shown.

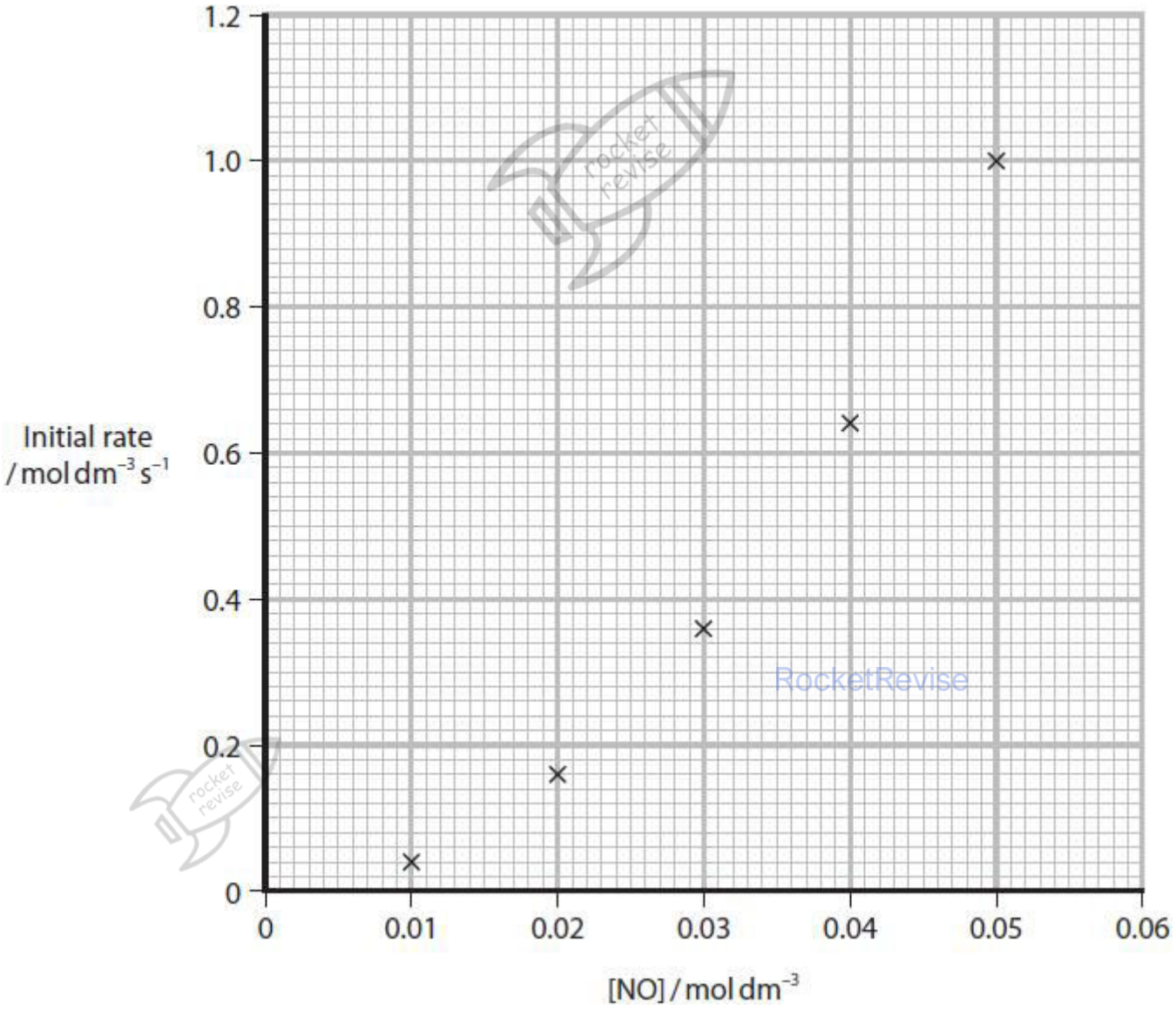
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Experiment	Initial [NO] / mol dm^{-3}	Initial [O_2] / mol dm^{-3}	Initial rate / mol dm^{-3} s^{-1}
1	0.010	0.050	0.040
2	0.020	0.050	0.160
3	0.030	0.050	0.360
4	0.040	0.050	0.641
5	0.050	0.050	1.001
6	0.020	0.025	0.080

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The data for experiments 1–5 were plotted on a graph.



(i) Draw a best-fit line on the graph. (1)

(ii) State how the graph shows that the reaction is **not** first order with respect to nitrogen monoxide. (1)

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(iii) Deduce the orders of reaction with respect to NO and O_2, using the data from experiments 1–6.

(2)

Order with respect to NO = ...
Order with respect to O_2 = ...

(iv) Write the rate equation for the reaction, using your answer to (a)(iii).

(1)

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(v) Calculate the rate constant for this reaction using the data from experiment 1 and your rate equation. Include units in your answer.

(2)

(b) The equilibrium constant, K_p, for the reaction at 298 K is 1.55×10^6 $atm^{-½}$.

State what this value of the equilibrium constant indicates about the position of the equilibrium. Justify your answer.

(2)

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(Total for question = 9 marks)

Q18.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

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The equation for the reaction between iron and steam is shown.

$$3Fe(s) + 4H_2O(g) \rightleftharpoons Fe_3O_4(s) + 4H_2(g) \qquad \Delta H = -151.2\,kJ\,mol^{-1}$$

(a) What is the equilibrium constant expression for this reaction?

(1)

☒ **A** $K_c = \dfrac{[Fe_3O_4] \times [H_2]^4}{[Fe]^3 \times [H_2O]^4}$

☐ **B** $K_c = \dfrac{[Fe_3O_4] \times [H_2]^4}{[H_2O]^4}$

☐ **C** $K_c = \dfrac{[H_2]^4}{[Fe]^3 \times [H_2O]^4}$

☐ **D** $K_c = \dfrac{[H_2]^4}{[H_2O]^4}$

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(b) What is the effect on the equilibrium constant for this reaction if small pieces of iron are replaced by iron powder and if the temperature is increased?

(1)

	Iron powder used	Temperature increased
☐ **A**	K_c increases	K_c increases
☐ **B**	K_c unchanged	K_c increases
☐ **C**	K_c increases	K_c decreases
☐ **D**	K_c unchanged	K_c decreases

(Total for question = 2 marks)

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Q19.

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Cracking reactions are used to obtain more useful compounds from the alkanes found in crude oil. An equation for the cracking of butane is shown.

$$C_4H_{10}(g) \rightarrow C_2H_6(g) + C_2H_4(g)$$

Thermodynamic data for the compounds in this reaction are given in the table.

	$C_4H_{10}(g)$	$C_2H_6(g)$	$C_2H_4(g)$
Standard molar entropy $S^{\ominus}$ / $JK^{-1}mol^{-1}$	310.1	229.5	219.5
Standard molar enthalpy change of formation $\Delta_f H^{\ominus}$ / $kJ\,mol^{-1}$	−126.5	−84.7	+52.2

(a) (i) Calculate the entropy change in the system, $\Delta S^{\ominus}_{system}$, for the cracking of butane.

Include a sign and units with your answer.

(2)

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(ii) Calculate the enthalpy change of reaction, $\Delta_r H^{\ominus}$, for the cracking of butane.

Include a sign and units with your answer.

(2)

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(iii) Calculate the entropy change in the surroundings, $\Delta S^{\ominus}_{surroundings}$ at 298 K for the cracking of butane, using your answer to (a)(ii).

Include a sign and units with your answer.

(2)

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(iv) Calculate the total entropy change, $\Delta S^{\ominus}_{\text{total}}$ total, at 298 K for the cracking of butane, using your answers to (a)(i) and (a)(iii).

Include a sign and units with your answer.

(1)

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(v) Calculate the temperature at which the cracking reaction becomes feasible.

(2)

(b) 5 mol of butane is cracked at 750 K. At equilibrium, 4.45 mol of ethene is formed and the total pressure is 1.20 atm.

$$C_4H_{10}(g) \rightleftharpoons C_2H_6(g) + C_2H_4(g)$$

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(i) Give the expression for the equilibrium constant, K_p, for this reaction.

(1)

(ii) Calculate the value of K_p, including units if required.

(5)

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(Total for question = 15 marks)

Q20.

This question is about the thermodynamics of the reaction:

$$N_2O_4(g) \rightarrow 2NO_2(g) \qquad \Delta H = +57.2\,kJ\,mol^{-1}.$$

Compound	Standard molar entropy at 298 K, $S^{\ominus}$ / J K^{-1} mol^{-1}	Standard molar enthalpy of formation at 298 K, $\Delta_f H^{\ominus}$ / kJ mol^{-1}	Colour
NO_2	+240.0	+33.2	brown
N_2O_4	+304.2		colourless

(a) Calculate the entropy change for the reaction, using the information in the table.

Include a sign and units in your answer.

(2)

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(b) Calculate the enthalpy change of formation, $\Delta_f H$, of N_2O_4 (g) at 298 K, using the information in the table and the enthalpy change of the reaction.

Include a sign and units in your answer.

(2)

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(c) Calculate the entropy change of the surroundings, $\Delta S_{surroundings}$, at 298 K.

Give your answer to an appropriate number of significant figures.
Include a sign and units in your answer.

(3)

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(d) (i) Use your answers to parts (a) and (c) to calculate the total entropy change, ΔS_{total}, for this reaction at 298 K.

(1)

(ii) This reaction can also be written as an equilibrium:

$$N_2O_4(g) \rightleftharpoons 2NO_2(g)$$

Calculate the temperature at which ΔS_{total} is zero for this equilibrium.

(2)

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(e) (i) Write the expression for the equilibrium constant, K_p, for this reaction, including the units, if any.

(2)

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(ii) In an experiment, 10 mol of $N_2O_4(g)$ was placed in a closed container at 50 °C. At equilibrium, 27% of the $N_2O_4(g)$ had dissociated, and the pressure in the container was 4.0 atm.
Calculate the value of K_p at 50 °C.

(4)

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(iii) The total pressure is doubled to 8.0 atm.
State the effect on K_p.

(1)

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.....

(iv) The total pressure is doubled to 8.0 atm at constant temperature.
Explain the change in the percentage dissociation of $N_2O_4(g)$ by considering the effect on the partial pressures of $NO_2(g)$ and $N_2O_4(g)$.

(3)

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(Total for question = 20 marks)

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Q21.

Iodine was dissolved in an organic solvent, trichloromethane, and the resulting solution added to an equal volume of deionised water. The mixture was then shaken, producing two immiscible solutions: iodine in water and iodine in trichloromethane.

At equilibrium, the equation for the reaction can be written as:

$$I_2(\text{trichloromethane}) \rightleftharpoons I_2(aq)$$

(a) What is the expression for this equilibrium constant, K_c?

(1)

☐ **A** $K_c = \dfrac{[I_2(\text{trichloromethane})]}{[I_2(aq)]}$

☐ **B** $K_c = \dfrac{(I_2(aq))}{(I_2(\text{trichloromethane}))}$

☐ **C** $K_c = \dfrac{I_2(aq)}{I_2(\text{trichloromethane})}$

☐ **D** $K_c = \dfrac{[I_2(aq)]}{[I_2(\text{trichloromethane})]}$

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(b) Which statement describes what is happening at equilibrium?

(1)

☐ **A** iodine molecules move from the water to the trichloromethane layer only

☐ **B** iodine molecules move from the trichloromethane to the water layer only

☐ **C** iodine molecules move from the water to the trichloromethane and from the trichloromethane to the water layer.

☐ **D** there is no movement of individual iodine molecules

(Total for question = 2 marks)

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Q22.

Calcium oxide is manufactured by heating limestone at 1000 °C for 30 minutes.

The equation for the reaction is:

$$CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g) \qquad \Delta_r H = +178\,\text{kJ mol}^{-1}$$

(a) The numerical value of the equilibrium constant for this reaction is increased by:

(1)

☐ **A** allowing the carbon dioxide to escape

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☒ **B** increasing the heating time
☒ **C** increasing the temperature
☒ **D** reducing the pressure

(b) Which is the correct expression for the equilibrium constant, K_c, for this reaction?

(1)

☒ **A** $K_c = [CO_2]$

☒ **B** $K_c = \frac{1}{[CO_2]}$

☒ **C** $K_c = \frac{[CaCO_3]}{[CaO][CO_2]}$

☒ **D** $K_c = \frac{[CaO][CO_2]}{[CaCO_3]}$

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(Total for question = 2 marks)

Q23.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

Ammonia is produced from nitrogen and hydrogen in the presence of a solid catalyst in an equilibrium reaction.

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g) \quad \Delta H = -92\,kJ\,mol^{-1}$$

(a) 10 mol of $N_2(g)$ is heated with 30 mol of $H_2(g)$ in a sealed container. At equilibrium, 8.0 mol of $NH_3(g)$ is present.

What is the **total** amount of gas, in mol, in the equilibrium mixture?

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(1)

☒ **A** 8.0
☒ **B** 20
☒ **C** 32
☒ **D** 48

(b) What is the expression for the equilibrium constant, K_p , for this reaction?

(1)

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☐ **A** $\frac{p(N_2) + p(H_2)^3}{p(NH_3)^2}$

☐ **B** $\frac{p(NH_3)^2}{p(N_2) + p(H_2)^3}$

☐ **C** $\frac{p(NH_3)^2}{p(N_2) \times p(H_2)^3}$

☐ **D** $\frac{p(N_2) \times p(H_2)^3}{p(NH_3)^2}$

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(c) Which change would increase the value of K_p for this reaction?

(1)

☐ **A** decreasing the temperature
☐ **B** decreasing the particle size of the catalyst
☐ **C** increasing the total pressure
☐ **D** increasing the concentration of hydrogen molecules

(Total for question = 3 marks)

Q24.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The equation for a reaction is shown.

$$A(g) + 2B(g) \rightleftharpoons 3C(s) + 4D(g)$$

(a) Some collisions between reactant molecules do not lead to the formation of products.

What is the best explanation for this?

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(1)

☐ **A** the reactant concentrations are too low
☐ **B** the collisions do not have sufficient energy
☐ **C** the reaction is at equilibrium
☐ **D** the molecules do not collide in the correct ratio

(b) What are the units of the equilibrium constant, K_p, for this reaction?

(1)

☐ **A** atm
☐ **B** atm^{-1}
☐ **C** atm^4
☐ **D** atm^{-4}

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(Total for question = 2 marks)

Q25.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

For all chemical reactions, the equilibrium constant, K, is related to the total entropy change, ΔS_{total}, by the expression shown.

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$$\Delta S_{total} = R \ln K$$

ΔS_{total} for a reaction is negative.

Which statement is true?

- ☐ **A** the value of K is less than 1
- ☐ **B** the value of K is greater than 1
- ☐ **C** the value of K is negative
- ☐ **D** the position of equilibrium lies to the right

(Total for question = 1 mark)

Q26.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

Magnesium nitrate decomposes on heating as shown.

$$Mg(NO_3)_2(s) \rightleftharpoons MgO(s) + 2NO_2(g) + \tfrac{1}{2}O_2(g)$$

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What is the expression for the equilibrium constant, K_p, for this reaction?

- ☐ **A** $K_p = p(NO_2)^2 p(O_2)^{\frac{1}{2}}$
- ☐ **B** $K_p = p(NO_2)^4 p(O_2)$
- ☐ **C** $K_p = \dfrac{p(MgO)\, p(NO_2)^2\, p(O_2)^{\frac{1}{2}}}{p(Mg(NO_3)_2)}$
- ☐ **D** $K_p = \dfrac{p(MgO)^2\, p(NO_2)^4\, p(O_2)}{p(Mg(NO_3)_2)^2}$

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(Total for question = 1 mark)

Q27.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Ammonia is produced by the reaction of nitrogen with hydrogen in the presence of an iron catalyst.

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g) \qquad \Delta H = -92\,kJ\,mol^{-1}$$

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(a) Which of the following statements about the catalyst is **not** correct?

(1)

- ☒ **A** it lowers the activation energy of the reaction
- ☒ **B** it has no effect on the equilibrium constant for the reaction
- ☒ **C** it alters the enthalpy change of the reaction
- ☒ **D** it reduces the energy cost of the reaction

(b) Which conditions favour the highest percentage of ammonia in an equilibrium mixture from identical amounts of nitrogen and hydrogen?

(1)

- ☒ **A** a temperature of 400 K and a pressure of 200 kPa
- ☒ **B** a temperature of 400 K and a pressure of 200 atm
- ☒ **C** a temperature of 400 °C and a pressure of 200 kPa
- ☒ **D** a temperature of 400 °C and a pressure of 200 atm

(Total for question = 2 marks)

Q28.

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Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒. and then mark your new answer with a cross ☒.

This question is about the equilibrium for the reaction shown.

$$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g) \quad \Delta H = -196\ kJ\ mol^{-1}$$

(a) Which is the expression for the equilibrium constant, K_p?

(1)

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☒ **A** $K_p = \dfrac{2(pSO_3)}{2(pSO_2)\times(pO_2)}$

☒ **B** $K_p = \dfrac{[SO_3]^2}{[SO_2]^2\times[O_2]}$

☒ **C** $K_p = \dfrac{(pSO_3)^2}{(pSO_2)^2\times(pO_2)}$

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☒ **D** $K_p = \dfrac{(pSO_2)^2\times(pO_2)}{(pSO_3)^2}$

(b) Which statement about this equilibrium is correct?

(1)

☒ **A** increasing the pressure increases the value of K_p
☒ **B** decreasing the pressure decreases the equilibrium yield of SO_3
☒ **C** adding a catalyst increases the equilibrium yield of SO_3
☒ **D** adding a catalyst increases the value of K_p

(c) What is the relationship between the equilibrium constant, K, and ΔS_{total}?

(1)

☒ **A** $\Delta S_{total} = R \ln K$
☒ **B** $\ln \Delta S_{total} = RK$
☒ **C** $\Delta S_{total} = R \log K$
☒ **D** $\log \Delta S_{total} = RK$

(Total for question = 3 marks)

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Q29.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

One of the stages in the manufacture of sulfuric acid involves the oxidation of sulfur dioxide to sulfur trioxide.
The equation for this reaction can be written in either of the two ways shown.

Equation **1** $2SO_2 + O_2 \rightleftharpoons 2SO_3$
Equation **2** $SO_2 + \frac{1}{2}O_2 \rightleftharpoons SO_3$

The value of K_p at 25°C using Equation **1** is 4.0×10^{24} atm^{-1}.

(a) What is the numerical value of K_p at 25°C using Equation **2**?

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(1)

- ☐ **A** 2.0×10^{12}
- ☐ **B** 2.0×10^{22}
- ☐ **C** 2.0×10^{24}
- ☐ **D** 4.0×10^{24}

(b) The relationship between K_p and K_c is shown.

$$K_c = \frac{K_p}{(R \times T)^{\Delta n}}$$

where T is the temperature in kelvin,
R is the gas constant 0.082 dm^3 atm mol^{-1} K^{-1} and
Δn = moles of product − moles of reactants.
What is the value of K_c at 25°C for the equilibrium using Equation **1**?

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(1)

- ☐ **A** 1.64×10^{23}
- ☐ **B** 1.95×10^{24}
- ☐ **C** 8.20×10^{24}
- ☐ **D** 9.77×10^{25}

(Total for question = 2 marks)

Q30.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

This question is about the effect of changes on a system at equilibrium.

(a) The equation for the reaction between hydrogen and iodine is shown.

$$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$$

What is the effect, if any, of an **increase** in pressure on the appearance of the equilibrium mixture?

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(1)

- ☐ **A** no change
- ☐ **B** the mixture becomes more purple
- ☐ **C** the mixture becomes more brown
- ☐ **D** the mixture becomes less brown

(b) The equation for the manufacture of ammonia by the Haber process is shown.

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g) \quad \Delta H^{\ominus} = -92.2\,kJ\,mol^{-1}$$

(1)

$K_p = 6.76 \times 10^5$ atm^{-2} at 298 K
Which change(s), if any, will result in an **increase** in the value of K_p?

- ☐ **A** an increase in pressure only
- ☐ **B** a decrease in temperature only

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- ☒ **C** an increase in pressure and an increase in temperature
- ☒ **D** no changes in pressure or temperature will affect the value of K_p

(c) The equation for the equilibrium between nitrogen dioxide and dinitrogen tetroxide is shown.

$$N_2O_4(g) \rightleftharpoons 2NO_2(g)$$

A change in conditions results in K_p increasing from 0.115 atm to 3.89 atm.
What can be deduced about the change in position of the equilibrium?

(1)

- ☒ **A** the equilibrium position has shifted towards the products
- ☒ **B** the equilibrium position has shifted towards the reactants
- ☒ **C** the equilibrium position is unchanged
- ☒ **D** there is almost complete conversion of reactants into products

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(Total for question = 3 marks)

Q31.

This question is about nitrosyl chloride, NOCl, which is a yellow gas.
Nitrosyl chloride decomposes into yellow−green chlorine, Cl_2, and colourless nitric oxide, NO, as shown.

$$2NOCl(g) \rightleftharpoons 2NO(g) + Cl_2(g)$$

(a) A mixture of nitrosyl chloride, chlorine and nitric oxide was placed in a sealed container and allowed to reach equilibrium.

Explain what you would expect to **see** when the pressure of the equilibrium mixture is increased?

(2)

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(b) 0.250 mol of nitrosyl chloride in a 500 cm^3 sealed container was heated to 200 °C. At equilibrium there was 0.016 mol of chlorine in the mixture.

(i) Write the expression for the equilibrium constant K_c.

(1)

(ii) Complete the table using the equilibrium equation and the data.

(2)

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Substance	NOCl	NO	Cl_2
Initial mol	0.250	0	0
Equilibrium mol			0.016
Concentration / mol dm^3			

(iii) Using your answers from (b)(i) and (b)(ii), calculate the equilibrium constant, K_c. Give your answer to an appropriate number of significant figures and include units.

(3)

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(iv) When the temperature is reduced, the value of the equilibrium constant, K_c, becomes smaller. Explain what can be deduced about the nature of the reaction.

(2)

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(Total for question = 10 marks)

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